



Q1 Find smallest and largest number

10 marks

WAP to find smallest and largest number in an array

input: 12 5 18 3 25 9

output:

Smallest number:3

Largest number:25

Your Code

```
1 # WAP to find the smallest and largest number in an array
2
3 arr = list(map(int, input("Enter the numbers: ").split()))
4
5 smallest = arr[0]
6 largest = arr[0]
7
8 for i in arr:
9     if i < smallest:
10         smallest = i
11     if i > largest:
12         largest = i
13
14 print("Smallest number:", smallest)
15 print("Largest number:", largest)
```

Input

12 5 18 3 25 9

Output

```
Enter the numbers: Smallest number:
3
Largest number: 25
```



Q2 Missing Number

10 marks

WAP to find missing number in an array of shuffled order

input: 4 2 1 6 5

output:3

Your Code

```
1 # WAP to find the missing number in an array
2
3 arr = list(map(int, input("Enter the numbers: ").split()))
4
5 n = max(arr)
6 total = n * (n + 1) // 2
7 missing = total - sum(arr)
8
9 print("Missing number:", missing)
```

Input

4 2 1 6 5

Output

```
Enter the numbers: Missing number:
3
```

SIMATS Online Programming Lab

PYTHON PROGRAMMING



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Q3 Second largest number

10 marks

WAP to find second largest number in an array

INPUT: 12 5 18 3 25 9

OUTPUT: 18

Your Code

```
1 # WAP to find the second largest number in an array
2
3 arr = list(map(int, input("Enter the numbers: ").split()))
4
5 arr = list(set(arr))    # Remove duplicate values
6 arr.sort()
7
8 print("Second largest number:", arr[-2])
```

Input

12 5 18 3 25 9

Output

```
'C:\PyComp\PyC72\python.exe' is
not recognized as an internal or
external command,
operable program or batch file.
```



Q6 Rotate an array by 'N' positions

10 marks

WAP to rotate an array by 'N' positions

INPUT: 2 5 6 1 3 4

OUTPUT: 6 1 3 4 2 5

Your Code

```
1 # Read array elements
2 arr = list(map(int, input().split()))
3
4 # Number of positions to rotate
5 n = 2
6
7 # Rotate array
8 arr = arr[n:] + arr[:n]
9
10 # Print rotated array
11 for i in arr:
12     print(i, end=" ")
```

Input

2 5 6 1 3 2

Output

6 1 3 2 2 5



Run



Save



Submit



Q10 Palindrome

10 marks

Write a python program to check whether the given string is a palindrome

input: malayalam

output: malayalam is a palindrome

input: saveetha

output: saveetha is not a palindrome

Your Code

```
1 # Read a string
2 s = input()
3
4 # Check palindrome
5 if s == s[::-1]:
6     print(s, "is a palindrome")
7 else:
8     print(s, "is not a palindrome")
```

Input

Malayalam

Output

Malayalam is not a palindrome

Visible Test Cases

Test Case 1



Q9 count vowels, consonants, digits, and spaces

10 marks

Write a python program to count vowels, consonants, digits, and spaces in a given sentence

INPUT: Hello World 123

OUTPUT:

Vowels:3

Consonants:7

Digits:3

Spaces:2

Your Code

```
1 # Read a sentence
2 text = input()
3
4 vowels = 0
5 consonants = 0
6 digits = 0
7 spaces = 0
8
9 for ch in text:
10     if ch.lower() in "aeiou":
11         vowels += 1
12     elif ch.isalpha():
13         consonants += 1
```

Input

HELLO WORLD:123

Output

```
Vowels: 3
Consonants: 7
Digits: 3
Spaces: 2
```



Q8 Print all 0's followed by non-zero without changing the order

10 marks

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```
1 # Read array elements
2 arr = list(map(int, input().split()))
3
4 # Separate zeros and non-zeros
5 zeros = []
6 non_zeros = []
7
8 for i in arr:
9     if i == 0:
10         zeros.append(i)
11     else:
12         non_zeros.append(i)
13
14 # Combine zeros first, then non-zeros
15 result = zeros + non_zeros
```

Input

1 0 4 0 2 5 0 7 0 6

Output

Security Error: Disallowed code pattern detected.

Visible Test Cases



Q7 Remove all duplicates in an array

10 marks

WAP to remove all duplicates in an array

INPUT: 2 4 6 2 7 8 3 4

OUTPUT: 2 4 6 7 8 3

Your Code

```
1 # Read array elements
2 arr = list(map(int, input().split()))
3
4 # Remove duplicates
5 result = []
6
7 for i in arr:
8     if i not in result:
9         result.append(i)
10
11 # Print result
12 for i in result:
13     print(i, end=" ")
```

Input

2 4 6 2 7 8 3 4

Output

2 4 6 7 8 3



Run



Save



Submit



Q5 Unique elements

10 marks

WAP to print all unique elements in an array

INPUT : 2 4 6 2 5 6 7

OUTPUT : 4 5 7

Your Code

```
1 # Read array elements
2 arr = list(map(int, input().split()))
3
4 # Print unique elements
5 for i in arr:
6     if arr.count(i) == 1:
7         print(i, end=" ")
```

Input

2 4 6 2 5 6 7

Output

4 5 7



Run



Save



Submit



Q4 Most repeated number

10 marks

WAP to print the most repeated number in an array

INPUT: 2 4 6 2 5 3 1 2

OUTPUT:2

Your Code

```
1 # Read array elements
2 arr = list(map(int, input().split()))
3
4 # Find the most repeated number
5 count = {}
6
7 for num in arr:
8     if num in count:
9         count[num] += 1
10    else:
11        count[num] = 1
12
13 most_repeated = max(count, key=count.get)
14
15 print(most_repeated)
```

Input

2 4 6 2 5 3 1 2

Output

2



Run



Save



Submit